Applying NLP for a Topic Modelling

Customer Feedback Analysis

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1. Introduction

PureGym receives a large volume of customer feedback across digital platforms such as Google Reviews and Trustpilot. While this feedback provides valuable insight into member experience and operational performance, its unstructured nature makes it difficult to systematically identify recurring issues and prioritise improvements at scale.

The objective of this project is to uncover dominant themes within negative customer reviews and translate these findings into actionable business insights. To achieve this, a combination of traditional topic modelling techniques and Large Language Model (LLM)-assisted analysis is employed. Using multiple analytical approaches enables both statistical robustness and qualitative depth, strengthening confidence in the findings.

By applying Natural Language Processing (NLP) techniques to customer reviews, this analysis identifies key drivers of dissatisfaction, highlights systemic operational challenges, and demonstrates how advanced language models can enhance interpretability beyond traditional methods.

2. Data Overview and Preparation

2.1 Data Sources

The analysis uses customer reviews collected from two independent platforms:

- **Google Reviews**
- **Trustpilot**

The datasets cover a twelve-month period and include reviews from 512 Google locations and 376 Trustpilot locations, with 310 locations common to both platforms. This overlap enables cross-platform validation and consistent comparison across shared gym locations. Only negative reviews were retained, as these provide the clearest signals of operational and service-related issues.

Table 1: Dataset summary (before and after filtering)

Source	Number of Reviews		No. of Unique Locations	No. of Negative Reviews
	Initial count	After removing empty reviews		
Google	23250	13898	512	2785
Trustpilot	16673	16673	376	3543

2.2 Data Cleaning and Preprocessing

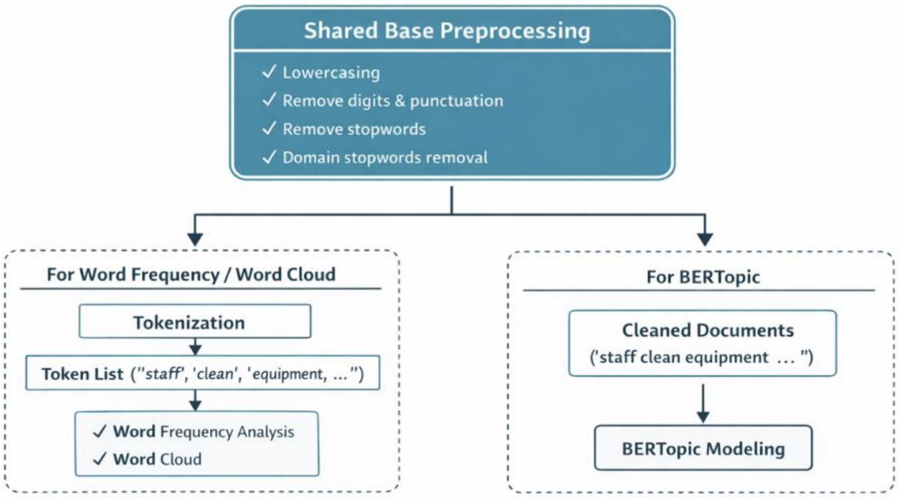
A shared preprocessing pipeline was applied across all analyses to ensure consistency. This included lowercasing text, removing numbers, punctuation, and special characters, eliminating standard and domain-specific stopwords, and normalising whitespace.

Two processed representations were created:

1. **Tokenised text**, used for word frequency analysis and word cloud visualisation
2. **Cleaned full documents**, preserved for topic modelling using BERTopic, LLM-based methods, and LDA

This design supports both lightweight exploratory analysis and deeper semantic modelling without compromising data integrity.

Figure 1: Summary of preprocessing steps and downstream usage



3. Analytical Approach

3.1 Exploratory Text Analysis

Exploratory analysis using word frequency distributions and word clouds was first conducted to gain an initial understanding of dominant vocabulary in customer feedback. Frequently occurring terms across both platforms included *equipment*, *staff*, *machines*, and *classes*, indicating a strong focus on facility usage and service experience.

When restricted to negative reviews, additional emphasis emerged around *dirty*, *shower*, *manager*, *people*, and *time*. While these techniques lack contextual depth, they confirm data quality and provide early indications of dissatisfaction drivers, informing subsequent modelling choices.

Figure 2: Top 10 words frequency bar charts

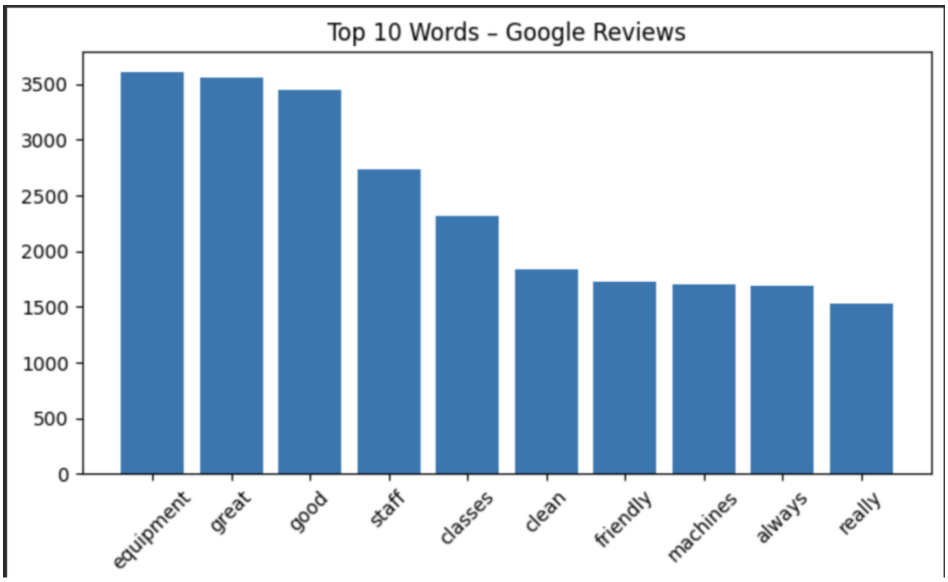


Figure 3: Word Cloud of Google Reviews

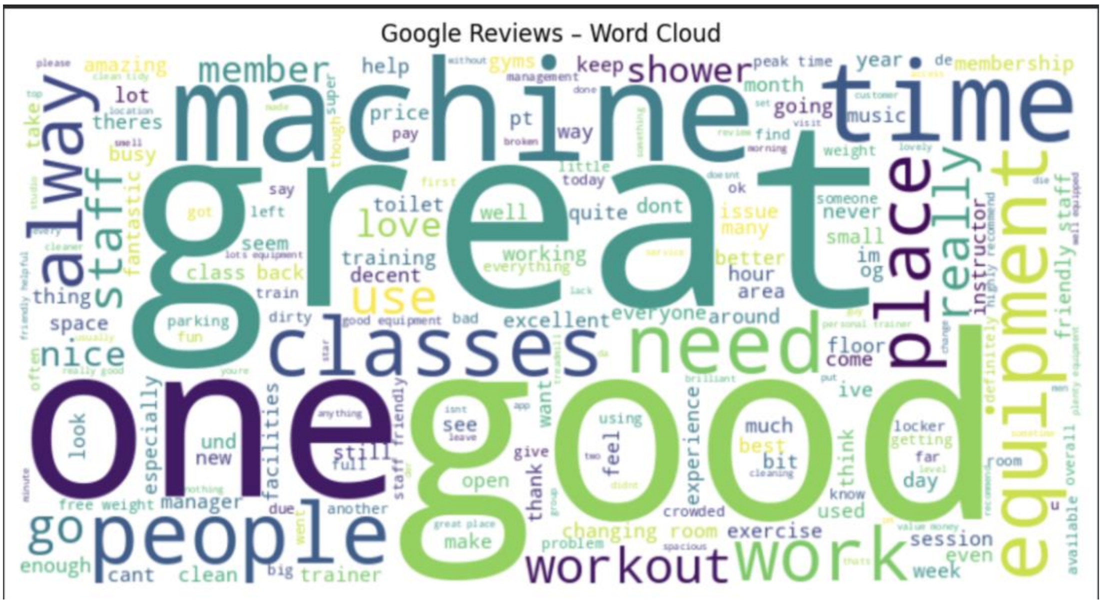


Figure 4: Word Cloud of Google Negative Reviews

3.2 Initial Topic Modelling with BERTopic

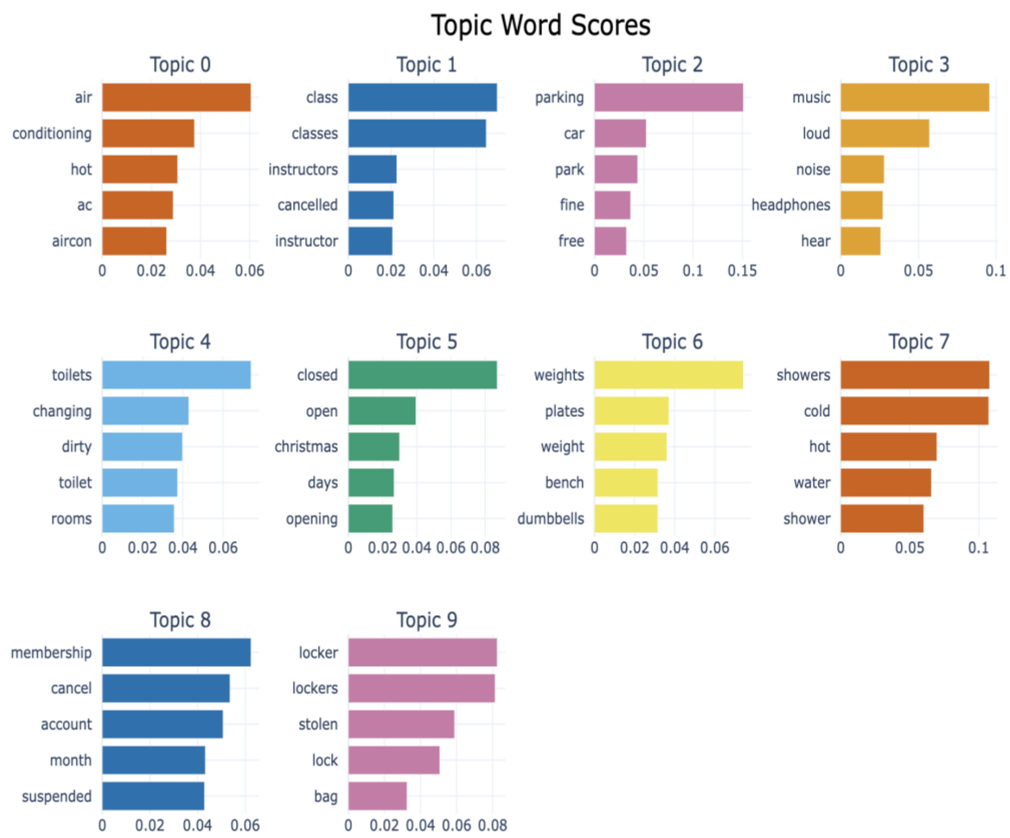
BERTopic was applied to negative reviews from locations common to both platforms to identify coherent themes. Leveraging transformer-based embeddings, BERTopic effectively clustered semantically similar complaints.

The resulting topics highlighted recurring issues such as:

- Poor ventilation and uncomfortable temperatures
- Cleanliness and hygiene concerns in changing rooms and toilets
- Overcrowding and limited equipment availability
- Class cancellations and membership management issues
- Parking constraints and restricted access hours

These findings indicate operational issues affecting the day-to-day gym experience rather than isolated incidents.

Figure 5: Topic bar chart (top words per topic)

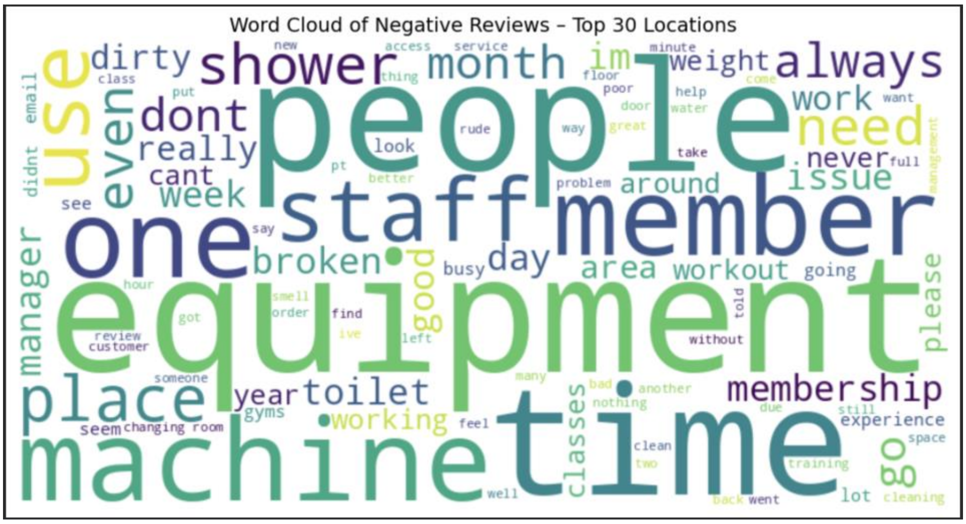


3.3 Topic Modelling with Top Locations

To assess whether dissatisfaction was location-specific, the top 20 locations with the highest volume of negative reviews were identified separately for Google and Trustpilot. Substantial overlap was observed, suggesting that complaints are concentrated in specific gyms rather than evenly distributed across the network.

Analysis of the top 30 locations revealed intensified versions of previously identified themes. Word clouds from these locations showed stronger emphasis on *overcrowding*, *equipment breakdowns*, *staff behaviour*, and *cleanliness*, indicating that high-complaint locations experience compounded operational challenges.

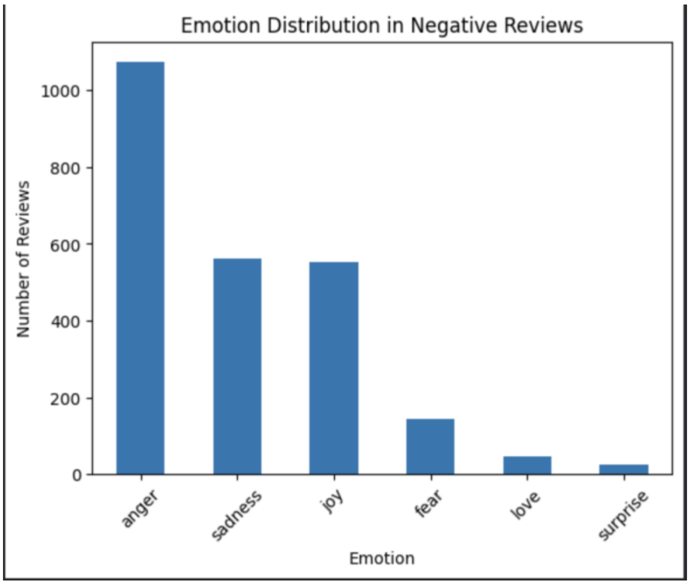
Figure 6: Word cloud for top 30 locations



3.4 Topic Modelling with Emotion Analysis

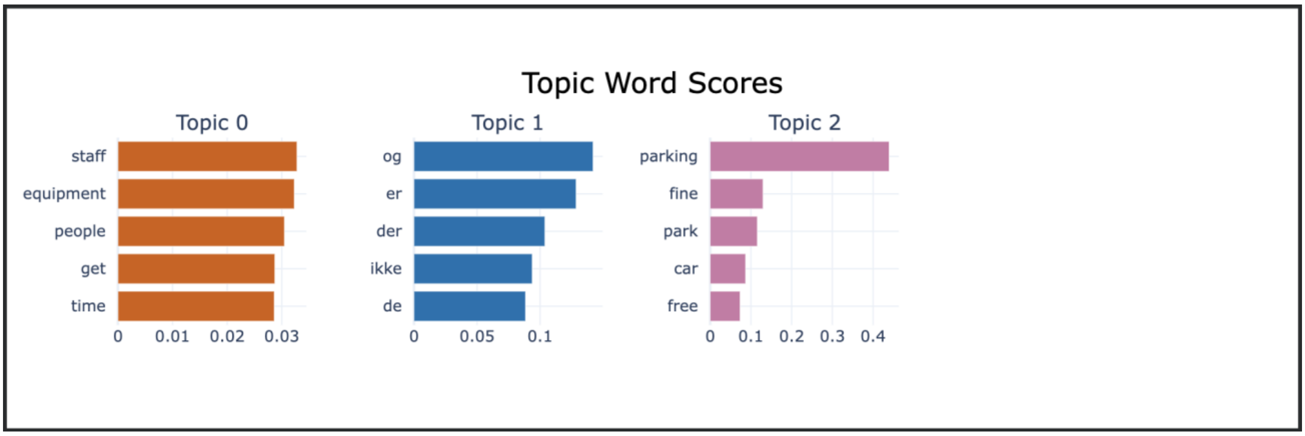
Emotion analysis using a BERT-based classifier revealed that *anger*, *frustration*, and *disappointment* dominate negative reviews. When focusing specifically on reviews classified as anger, the issues narrowed to recurring unresolved problems, including rude staff interactions, chronic overcrowding, and neglected maintenance.

Figure 7: Emotion distribution bar chart



Applying BERTopic to anger-filtered reviews further refined these themes, demonstrating that emotional intensity is often driven by repeated failures rather than one-off inconveniences.

Figure 8: BERTopic - bar chart visual of Angry Reviews

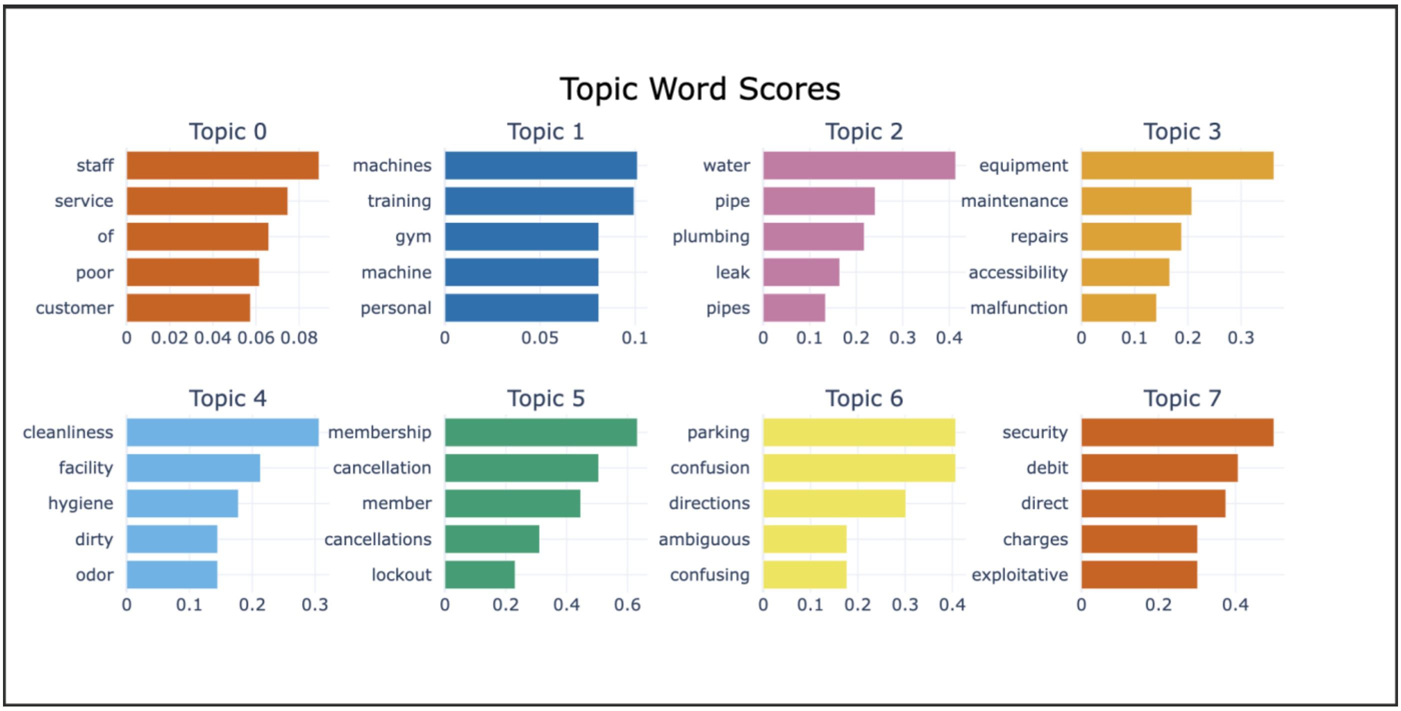


3.5 LLM-Assisted Topic Extraction and Insight Generation

To complement statistical models, an LLM-based pipeline was implemented to extract abstract topic phrases and generate business-oriented insights. Initial experiments with FLAN-T5 produced summarisation rather than meaningful topic extraction, while Falcon-7B-Instruct could not be deployed due to memory constraints. Phi-4-mini executed locally but proved difficult to integrate and reproduce.

As a result, **Qwen2.5-3B-Instruct** was selected due to its efficiency, prompt adherence, and compatibility with GPU-accelerated Colab environments. The model extracted concise topic phrases from a subset of negative reviews, which were then aggregated and passed through BERTopic for clustering. This LLM-assisted approach produced cleaner, more interpretable themes and enabled the generation of actionable recommendations aligned with business decision-making.

Figure 9: BERTopic - bar chart visual of LLM extracted topics



3.6 Topic Modelling with LDA (Gensim)

Latent Dirichlet Allocation (LDA) was implemented as a probabilistic baseline with 10 topics. LDA surfaced consistent themes related to equipment issues, showers and water supply, staff behaviour, and parking constraints. However, some topic overlap was observed, reflecting limitations when modelling short, informal review text.

Figure 10: Top terms per LDA topic

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Topic 0: ['air', 'equipment', 'like', 'hot', 'conditioning', 'working', 'good', 'con', 'really', 'work']
Topic 1: ['parking', 'closed', 'email', 'membership', 'get', 'hours', 'car', 'day', 'time', 'staff']
Topic 2: ['class', 'classes', 'dont', 'membership', 'customer', 'help', 'manager', 'service', 'pt', 'cancelled']
Topic 3: ['equipment', 'people', 'get', 'time', 'staff', 'one', 'machines', 'busy', 'machine', 'even']
Topic 4: ['day', 'showers', 'changing', 'pass', 'shower', 'cold', 'room', 'rooms', 'equipment', 'work']
Topic 5: ['water', 'showers', 'dont', 'months', 'cold', 'like', 'one', 'machine', 'shower', 'hot']
Topic 6: ['weights', 'membership', 'get', 'would', 'pin', 'experience', 'back', 'people', 'one', 'staff']
Topic 7: ['dirty', 'cleaning', 'changing', 'toilets', 'rooms', 'always', 'toilet', 'clean', 'broken', 'equipment']
Topic 8: ['machines', 'equipment', 'weights', 'area', 'people', 'space', 'one', 'use', 'small', 'music']
Topic 9: ['staff', 'equipment', 'member', 'manager', 'one', 'use', 'would', 'good', 'dont', 'members']
```

4. Results and Key Insights

4.1 Dominant Customer Pain Points

Across all modelling techniques, the following themes consistently emerged:

- Overcrowding and capacity constraints
- Poor hygiene and cleanliness standards
- Equipment availability and maintenance delays
- Negative staff interactions and communication issues
- Facility-related issues (showers, water, ventilation)
- Parking, billing, and access limitations

The recurrence of these themes across platforms and methods suggests systemic operational issues rather than isolated failures.

4.2 Comparison of Modelling Techniques

- **LDA** provided an interpretable baseline but struggled with topic overlap.
- **BERTopic** offered stronger semantic coherence and visual exploration.
- **LLM-assisted approach** produced the clearest abstractions and most business-ready insights.

Combining these methods increased analytical confidence and reduced reliance on any single technique.

5. Actionable Recommendations

Based on the consolidated findings, the following actions are recommended:

- Implement capacity monitoring and incentivise off-peak usage
- Strengthen hygiene and proactive equipment maintenance protocols
- Enhance staff training, communication standards, and accountability
- Prioritise repairs for showers and ventilation systems
- Improve clarity around parking, billing, and cancellation policies

6. Limitations and Considerations

- Computational constraints limited full-scale LLM execution
- Short and multilingual reviews reduced topic coherence in some cases
- Topic models inherently simplify complex customer experiences

Despite these limitations, strong alignment across methods supports the reliability of the conclusions.

7. Conclusion

This project demonstrates the value of combining traditional topic modelling with LLM-assisted analysis to extract actionable insights from unstructured customer feedback. While each method has limitations, their combined application provides a robust and interpretable understanding of customer dissatisfaction drivers.

The findings highlight clear opportunities for targeted operational improvements that can enhance customer satisfaction, retention, and brand perception. Integrating such analytical frameworks into ongoing feedback monitoring can support proactive, data-driven decision-making.